Unit II Question Bank

Unit II - Combinational Logic Design

Half Adder, Full Adder:

Q1: Design Full-adder circuit using two Half-adders. In diagram use block diagram of half-adder.

Half Subtractor, Full Subtractor:

Q1: Design Full-subtractor circuit using two Half-subtractors. In diagram use block diagram of half-subtractors.

Binary Adder (IC 7483): BCD adder:

Q1: Design single digit BCD adder using 4-bit binary adder IC7483 and explain two examples of BCD addition.

Look ahead carry generator:

Q1: Write a short note on look ahead carry generator also explain its importance.

Code Converters: BCD, Excess-3, Gray code, Binary Code:

Q1: Design BCD to Excess-3 code converter using minimum number of gates.

Q2: Design BCD to Excess-3 code converter using minimum Number of NAND gates.

Q3: Design Excess-3 to BCD code converter using minimum number of gates.

Q4: Design Excess-3 to BCD code converter using minimum Number of NAND gates.

Q5: Design 4-bit Binary to Gray code converter using minimum number of gates.

Q6: Design 4-bit Gray to Binary code converter using minimum number of gates.

Multiplexers:

Q1: Implement the expression using 8:1 multiplexer:

$f(A, B, C, D) = \sum m\ (0, 2, 3, 6, 8, 9, 12, 14)$

Demultiplexers:

Q1: Implement the following multi-output combinational logic circuit using 4-Line-to-16-Line decoder.

$F1 = \sum m(1, 2, 4, 7, 8, 11, 12)$

$F2 = \sum m(2, 3, 10, 11, 14)$

$F3 = \sum m(11, 12, 14, 15)$

$F4 = \sum m(3, 7, 9, 13)$

Comparators (2 bit):

Q1: Design 2-bit comparator with minimum number of gates.